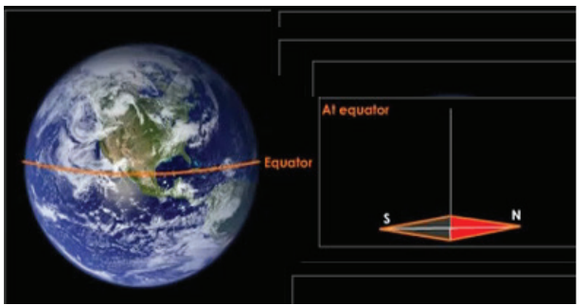


Figure 3.6: Needle of magnetic compass rests horizontally at an angle of dip – at magnetic equator

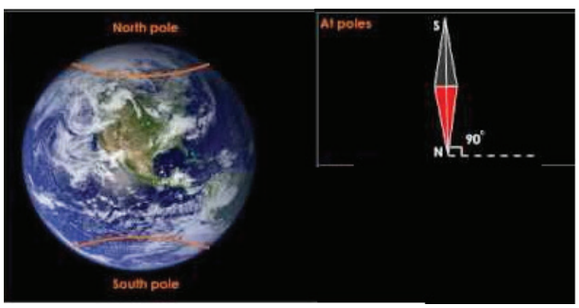


Figure 3.7: Needle of magnetic compass rests vertically at an angle of dip – at magnetic poles

of magnetic compass rests vertically at an angle of dip, I = 90° as shown in Figure 3.7. Hence,

$B_H = 0$

$B_V = B_E$

This implies that the vertical component is maximum at poles and horizontal component is zero at poles.

EXAMPLE 3.1

The horizontal component and vertical components of Earth's magnetic field at a place are 0.15 *G* and 0.26 *G* respectively. Calculate the angle of dip and resultant magnetic field.

Solution:

$B_H = 0.15$ G and $B_V = 0.26$ G

$$\tan I = \frac{0.26}{0.15} \Rightarrow I = \tan^{-1}(1.732) = 60^\circ$$

The resultant magnetic field of the Earth is

$$B = \sqrt{B_H^2 + B_V^2} = 0.3G$$

3.1.2 Basic properties of magnets

Some basic terminologies and properties used in describing bar magnet.

(a) Magnetic dipole moment

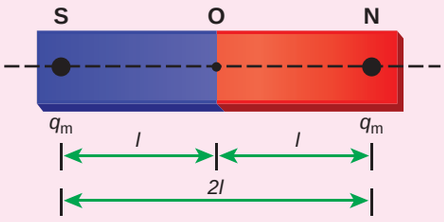


Figure 3.8 A bar magnet

Consider a bar magnet as shown in Figure 3.8. Let q_m be the pole strength (it is also called as magnetic charge) of the magnetic pole and let l be the distance between the geometrical center of bar magnet O and one end of the pole. **The magnetic dipole moment is defined as the product of its pole strength and magnetic length**. It is a vector quantity, denoted by $\vec{p}_m$.

$$\vec{p}_m = q_m \vec{d} \qquad (3.4)$$

where $\vec{d}$ is the vector drawn from south pole to north pole and its magnitude $|\vec{d}| = 2l$.

The magnitude of magnetic dipole moment is $p_m = 2q_m l$

UNIT-3(XII-Physics_Vol-1).indd 132 04-03-2019 10:33:18